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Whole-flint corn grain or tropical grass hay free choice in the diet of dairy calves

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ABSTRACT

Tropical grass hay feeding is related to improved ruminal health; however, it may decrease energy intake. On the other hand, whole-flint corn grain may be an alternative fiber source in the diet of dairy calves. Forty-two Holstein calves were used in a randomized block design, considering sex, birth date, and weight at 21 d of age, when the supply of whole-flint corn grain or tropical grass hay started. Three component-fed solid diets were compared: (1) starter concentrate only during the preweaning and starter concentrate with free choice of chopped Tifton-85 hay postweaning (SC), (2) starter concentrate with free choice of chopped Tifton-85 hay pre- and postweaning (SCH), and (3) starter concentrate with free choice of whole-flint corn grain pre- and postweaning (SCW). The animals were evaluated from 21 to 84 d of age. Calves were managed equally during the first 21 d, fed with 6 L/d of whole milk and a commercial starter concentrate (46% nonfiber carbohydrates, small particles, and pelleted) ad libitum. After that, milk feeding was reduced to 4 L/d until gradual weaning at 56 d of age. At 56 d of age, 4 animals per treatment were randomly chosen to be slaughtered for digestive tract weight evaluation and to collect tissue for histological analysis of the ruminal wall, duodenum, and cecum, whereas the other 30 animals were weaned and evaluated for a further 22 d when the SC diet also received hay ad libitum. Feed intake was measured daily. Weight gain and metabolic indicators of intermediate metabolism were evaluated weekly. Ruminal fluid was collected at wk 6, 8, 10, and 12 of age. The SCH diet increased the total and starter dry matter intake, and consequently, the average daily gain and body weight at 56 d of age. The SCW diet promoted an increase in propionate and decreased acetate-to-propionate ratio. Morphometric variables were affected by the SCH diet. The postweaning performance was unaffected by solid diets; however, the SCW diet decreased ruminal and fecal pH. Feeding hay, starting at 21 d of age, can stimulate early solid diet intake, promoting better performance and ruminal and intestinal development, when a highly fermentable and small particle pelleted starter is fed.
Key words: papillae, villus, pH, postweaning

INTRODUCTION

The nature of solid ingredients, amount ingested, carbohydrate inclusion levels, grain processing (Lesmeister et al., 2004; Porter et al., 2007; Khan et al., 2008), and forage (Suárez et al., 2007; Castells et al., 2013) affect ruminal fermentation patterns and alter ruminal development. Increased ruminal fermentation products, such as VFA, promote ruminal epithelium development, providing a more extensive exchange area between the rumen content and bloodstream (Sander et al., 1959). With the developed rumen, the animals can ferment solid diets, and their epithelium is able to absorb and metabolize VFA, making them available as an energy source for calves.

Many commercial starters are in a pelleted physical presentation and are formulated to have high NFC levels to promote the ruminal development and performance of calves. It is common to find commercial starters with inadequate fiber levels, resulting in negative effects, such as low solid feed intake and reduced ruminal health (Khan et al., 2016). The finely ground corn contained in the pelleted starter is digested faster in the rumen and more completely over the total digestive tract than coarsely ground corn (Rémond et al., 2004). In addition, soft pellets break up during handling or while exiting the pellet mill, leading to a solid diet that may contain large amounts of fines (Bateman et al., 2009). Solid diets with reduced particle size can decrease starter intake and lead to ruminal parakeratosis

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when supplied to calves in the initial phase (Greenwood et al., 1997).

Nonprocessed corn provides a slow passage rate and a slow starch fermentation, which reduces the excessive production of lactate in the rumen (Britton and Stock, 1987). Whole-flint corn grain (*Zea mays* ssp. *Indentura*) is rich in a vitreous endosperm, characterized as compact, and having a dense protein matrix, with less available starch (Corona et al., 2006). This corn is the most widely grown cereal grain in South America and the primary grain used in diets for ruminants in Brazil (Correa et al., 2002). The mean particle size and the physically effective neutral detergent fiber (**peNDF**) indicate a possible potential of whole-flint corn grain to promote chewing and prevent ruminal acidosis (Yang and Beauchemin, 2006). When calves are fed low-fiber pelleted starter, the whole-flint corn grain in the diet may be an alternative to stimulate mechanical chewing and rumination without reducing energy intake, while facilitating the management of a solid diet.

On the other hand, forage supplementation increases and stabilizes the ruminal pH, which is important to ensure ruminal health and guarantee solid diet intake for a calf with a developing rumen. However, studies have shown that the provision of high-quality forage ad libitum before weaning, such as alfalfa hay with high CP content 17 to 20% of DM and low NDF content 47% of DM, resulted in a substitution effect, decreasing concentrate and consequently total nutrient intake (Hill et al., 2008; Quigley et al., 1992). Further studies are needed to evaluate the perennial tropical forages effects on young calves' diets. *Cynodon dactylon*, especially the cultivar Tifton-85 (*Cynodon* spp.), has shown promising results for adult ruminant feeding due to its characteristics of high productivity, acceptance by animals, and good nutritional composition with 14.37% CP, 79.67% NDF, and 58.40 TDN (Ribeiro et al., 2001).

This study evaluated Holstein calves fed a commercial pelleted starter concentrate with or without whole-flint corn grain or tropical grass hay free choice in the solid diet, both pre- and postweaning period, investigating effects in the short and medium terms. We have hypothesized that (1) feeding tropical grass hay free choice, starting during the preweaning period, may result in higher NDF intake without reducing performance, given the reduced risk for ruminal acidosis of this diet relative to feeding a commercial pelleted starter only; and (2) feeding whole-flint corn grain free choice, starting during the preweaning period, may result in a higher intake of nutrients, improving performance and ruminal development, but with increased ruminal health risk, as compared with feeding a commercial starter exclusively or combined with tropical grass hay free choice. The objective of our study was to investigate the supply of whole-flint corn grain, since preweaning, and tropical grass hay, before and after weaning, free choice associated with highly fermentable and small particle pelleted starter on performance, skeletal growth, ruminal fermentation, blood metabolites, and ruminal and intestinal development of dairy calves.

MATERIALS AND METHODS

This study was conducted at the calf facilities of the Department of Animal Science, "Luiz de Queiroz" College of Agriculture, University of São Paulo, located in Piracicaba, São Paulo, Brazil. The trial period was from December 2017 to July 2018. During this period, the average temperature was 23°C, with a maximum of 34.5°C and a minimum of 5.2°C. The mean relative humidity during the study period was 74%, and the average rainfall was 89.5 mm/mo. All animal procedures followed the protocol approved by the Animal Care and Use Committee of the Luiz de Queiroz College of Agriculture, University of Sao Paulo (Piracicaba, Sao Paulo, Brazil; protocol no. 2018.5.587.11.3).

Animals, Experimental Design, and Treatments

Forty-two Holstein newborn calves (33 males: 39.47 ± 0.850 kg; and 9 females: 35.43 ± 0.855 kg), were used in a randomized complete block design. Calves were blocked by sex, birth date, and weight at 21 d of age when the supply of whole corn grain or hay started. The colostrum quality was measured using a colostrometer, and only high-quality colostrum (>60 mg/L of IgG) was fed to the newborn calves, in a volume to correspond to 10% of their birth weight within 6 h after birth. Blood samples were obtained by venipuncture from the jugular vein at 48 h after the first colostrum intake, and total serum protein (**TSP**) was determined as an indicator of passive transfer of immunity using a digital handheld refractometer (ITREF 200, Instrutemp, São Paulo, SP, Brazil). The colostrum protocol resulted in a mean TSP of 5.8 ± 0.24 g/dL, suggesting no failure of passive immune transfer (Elsohaby et al., 2019).

The animals were evaluated from 21 to 84 d of age. Calves were managed equally during the first 21 d when they were fed with 6 L/d of whole milk, supplied in open buckets, divided into 2 meals (7 and 17 h), and the commercial pelleted starter concentrate (Ração Bezerra Ag Milk Agroceres Multimix Nutrição Animal Ltda., Rio Claro, SP, Brazil). After this period, the volume of milk was reduced to 4 L/d of whole milk (12.7% solids, 3.96% fat, 3.30% protein, and 4.44%

lactose) until weaning started at 56 d of age. Three different component-fed solid diets were evaluated: (1) starter concentrate only, during the preweaning period, and starter concentrate with free choice of chopped Tifton-85 hay during the postweaning period (**SC**); (2) starter concentrate with free choice of chopped Tifton-85 hay (17.41 mm), during the pre- and postweaning period (**SCH**); and (3) starter concentrate with free choice of whole-flint corn grain during the pre- and postweaning period (**SCW**). The chemical composition and ingredients of the diets are shown in Table 1. The animals were gradually weaned (reducing 0.5 L per day), regardless of the solid diet intake, starting at 56 d until the end of 62 d. After the completion of weaning, the animals were evaluated for a further period of 22 d. Data were split into a preweaning (1–56 d) and postweaning (57–84 d) period, meaning that the postweaning period included the weaning process. After 56 d of age, SC calves also received tropical grass hay ad libitum, which is a common feeding strategy adopted by producers in Brazil. In addition, after 56 d, concentrate starter offered was maximized at 2 kg/d, but tropical grass hay and whole-flint corn grain were fed for ad libitum consumption.

Calves were housed in individual shelters (1.35 m height, 1 m width, and 1.45 m depth) and contained by a chain belt attached to a thin chain (2 m long), allowing an adequate walking area, but with no physical contact with other animals. Shelters were distributed in a *Paspalum notatum* grass field frequently trimmed to ground level, so no grass was available for grazing. Free access to clean water was provided throughout the study in individual buckets. Calves had free access to the starter concentrate and tropical grass hay or whole-flint corn grain in 2 separate feed bunks (component diet). Starter delivery was done just after morning milk feeding and was available until the next day.

Feed samples were collected monthly (n = 7) to determine the offered feed's chemical composition. Feed samples were dried (MA035, Marconi, Piracicaba, São Paulo, Brazil) at 55°C for 72 h and ground through a 1-mm screen Wiley mill (Marconi). The DM was determined by drying the samples in an oven at 105°C for 24 h (AOAC International, 2002; method 925.40) and ash by incinerating the samples in a muffle furnace at 550°C for 4 h (AOAC International, 2002; method 942.05). Sequential detergent fiber analyses were used to determine NDF (Van Soest et al., 1991) and ADF

Table 1. Ingredients and chemical composition of experimental diets

Item	Starter	Grass hay	Whole corn
Ingredient, % DM			
Ground corn	32.30	—	—
Soybean meal	29.20	—	—
Wheat meal	15.00	—	—
Expeller soybean	8.5	—	—
Pregelatinized corn	10.0	—	—
Mineral and vitamin supplement[2]	5.0	—	—
Chemical composition of diets, %			
DM	89.3	82.5	87.0
CP	24.7	12.7	9.3
ADF	5.5	32.5	3.4
NDF	13.9	74.7	12.4
Ash	9.7	6.2	1.2
Ether extract	5.2	2.5	5.4
NFC	46.6	3.8	71.8
Particle size distribution, %			
19 mm	0.0	60.71	0.0
8 mm	8.86	23.81	81.12
4 mm	81.46	8.33	18.45
1.18 mm	5.06	4.76	0.43
Bottom pan	4.64	2.38	0.0
MPS,[3] mm	5.48	17.41	10.60
peNDF[4] > 8.0 mm, %	1.17	61.64	10.06

[1]Starter = starter concentrate; grass hay = chopped hay; whole corn = whole-flint corn grain.

[2]Mineral and vitamin supplement, composition: 20% Ca; 6.5% P; 650 mg/kg F; 4% In; 1% K; 7% Mg; 0.7% S; 25 mg/kg Co; 800 mg/kg Cu; 20 mg/kg Cr; 40 mg/kg I; 1,400 mg/kg Fe; 1,500 mg/kg Mn; 18 mg/kg Si; 3,200 mg/kg Zn; 140,000 IU/kg vitamin A; 50,000 IU/kg vitamin D_3; 1,500 IU/kg vitamin E; 250 mg/kg vitamin B_1, 250,000 mg/kg vitamin B_2; 250 mg/kg vitamin B_6; 250 mg/kg vitamin B_{12}; 400 mg/kg niacin; 500 mg/kg pantothenic acid; 20 mg/kg folic acid; 10 mg/kg biotin; 800 mg/kg butylated hydroxytoluene, and 900 mg/kg sodium monensin.

[3]Mean particle size.

[4]Physically effective neutral detergent fiber.

(Goering and Van Soest, 1970) concentrations in an Ankom 2000 fiber analyzer (Ankom Tech. Corp., Fairport, NY). Heat-stable α-amylase and sodium sulfite were included in the NDF analysis. Total nitrogen concentration was determined using the Leco TruMac N apparatus (Leco Corporation, St. Joseph, MI; AOAC International, 2002; method 968.06). The CP was calculated by multiplying the total nitrogen by 6.25. The ether extract concentration was determined using petroleum ether (AOAC International, 2002; method 920.39). The NFC of the diets were estimated according to the following equation: NFC (%) = 100% − (% NDF + % CP + % fat + % ash), according to Mertens (1997).

Samples for particle size analysis were separated using the 4-screen Penn State Particle Separator (long: >19 mm; medium: between 8 and 19 mm; short: between 8 and 1.18 mm; and fine: <1.18 mm) and mean particle size was calculated. The peNDF was calculated considering particles >8 mm, which were retained in the top 2 screens of the Penn State Particle Separator (Yang and Beauchemin, 2006).

Samples of bulk milk were collected monthly (n = 7) during the overall experiment period for TS analyzed by Fourier-transform infrared spectroscopy (Lefier et al., 1996). A TS concentration of milk samples was used for feed efficiency (**FE**) calculation.

Measurements, Sampling, and Analyses

The solid daily intake (starter, hay and, whole corn grain) was measured every morning by weighing orts separately. After weighing refusals, feeding was calculated considering the previous day's intake with an increase of at least 150 g. Calves were weighed at birth, weekly until wk 8 and then at wk 10 and 12, using a mechanical scale (ICS-300, Coimma Ltda., Dracena, SP, Brazil), always before the morning feeding. Average daily gain and feed efficiency (kg of BW gain/kg of total DMI) were calculated for preweaning (21–56 d) and postweaning (57–84 d) periods. Body measures, such as withers height, heart girth, and hip width, were recorded at 6 and 8 wk (preweaning) and 10 and 12 wk (postweaning) according to the method described by Khan et al. (2008).

Blood samples were obtained weekly by venipuncture of the jugular vein into an evacuated tube containing sodium fluoride as antiglycolytic and potassium EDTA as anticoagulant (Vacuette do Brasil, Campinas, SP, Brazil), and evacuated tube containing clot activator and gel (Vacuette do Brasil), always 2 h after the morning feeding. Tubes were immediately placed in ice and transported to the laboratory. Samples were centrifuged at 2,000 × g for 15 min at 4°C to separate the serum and plasma, which were then stored at −10°C for subsequent analysis. The metabolic indicators were determined in the Automatic System for Biochemistry–Model SBA-200 (CELM, Barueri, SP, Brazil). Commercial kits from Labtest Diagnostica SA (Lagoa Santa, MG, Brazil) were used for the determination of plasma glucose (Ref. 133–1/500), TSP (Ref. 99), urea (Ref. 104), and L-lactate (Ref. 116). Concentration of BHB determination was carried out using the RANBUT enzyme kit (Ref. RB1007, Randox Laboratories–Life Sciences Ltd., Crumlin, UK).

Ruminal fluid samples were obtained 2 h after the morning feeding at wk 6, 8 and 10, 12, using a flexible esophageal tube (150 cm long, 1.3-cm internal diameter, and 0.2-cm wall thickness) connected to the vacuum pump (model TE-0581, Tecnal Ltda., Piracicaba, SP, Brazil). The samples were collected, discarding the initial portion. Visual appraisal of ruminal fluid was performed for screening of saliva contamination according to the method described by Terré et al. (2013a). After filtration in a cotton cloth, 100 mL of liquid was obtained, the pH was determined (model Tec-5, Tecnal Ltda.), and the samples were stored in a freezer (−10°C) for later analysis of VFA according to Ferreira et al. (2016) and ammonia nitrogen (NH_3-N) according to de Paula et al. (2017). The variable C2: (C3 + C4) was calculated according to Poczynek et al. (2019) through the ratio of acetic acid (**C2**) to the sum of propionic (**C3**) and butyric acid (**C4**), the essential acids related to ruminal epithelium development.

The fecal score was monitored daily as described by Larson et al. (1977) regarding the fluidity of feces, with (1) normal and firm; (2) loose but with general healthy aspect; (3) very loose, no watery separation; and (4) watery. At wk 6, 8 and 10, 12, fecal pH was obtained 2 and 7 h after the morning feeding, using 4 g of feces plus 4 mL of deionized water, in the digital pH meter (Modelo tec-5, Tecnal Ltda.), according to Channon et al. (2004).

Weight of Organs and Parameter Morphometrics

At 56 d of age, 4 blocks (4 animals per treatment) were randomly selected to be slaughtered. The content of the digestive tract was washed out, and the forestomach compartments divided into reticulum-rumen, omasum, and abomasum. The maximum capacity of the reticulum-rumen was determined with the aid of moorings at the end of this compartment, which was filled with water to its maximum capacity and the volume measured. After removal of excess water from the tissues, weights of the forestomach compartments, small intestine, and large intestine were recorded. Subsequently, a 1-cm^2 section was collected from the rumen

ventral sac for measured histological assessment. The number of papillae per square centimeter of the ruminal wall and histological measurements were determined as described by Lesmeister et al. (2004). Histological sections were stained with hematoxylin and eosin, embedded in paraffin wax, and sectioned (Odongo et al., 2006). Histological measurements, such as papillae height, papillae width, papillae surface area, and keratinized layer thickness (**KLT**), were determined in 4 papillae per animal using a computer-aided light microscope image analysis (Lesmeister et al., 2004).

The ruminal wall absorptive surface area (**ASA**) in square centimeters was calculated as follows: 1 + (the number of papillae × papillae surface area) − (the number of papillae × 0.002), where 1 represents the 1-cm^2 fragment collected and 0.002 is the estimated basal area of papillae in square centimeters (Barducci et al., 2019). The papillae area expressed as a percentage of ASA was calculated to understand how much of the ASA is represented by papillae as follows: (number of papillae × papillae surface area)/ASA × 100 (adapted from Pereira et al., 2020).

The small and large intestine compartments were also separated, emptied, and washed for sampling. Samples were collected in the initial portion of the duodenum, 1 cm^2 from the pylorus, and 1 cm^2 from the cecum (middle portion), and stored in 4% buffered paraformaldehyde to perform the histological analyses. For duodenum morphology, the villus lengths, depth of crypts, and the number of goblet cells were measured (Kehoe et al., 2019). For cecum morphology, cecum tissue was embedded in paraffin blocks, sectioned at 6 μm, and stained with hematoxylin and eosin. Crypt depths were measured for each intestinal section, and means were calculated per segment for each calf. The number of goblet cells was measured inside from the crypts so that the mean was considered (Pereira et al., 2020). In addition, the liver, kidneys, and spleen were removed, washed with water, and after removal of excess water, weights were recorded.

Statistical Analysis

Statistical analyses were performed using the MIXED procedure of SAS (version 9.0; SAS Institute Inc., Cary, NC). All data were tested for normal distribution by the Shapiro-Wilk test, and homogeneity of the variances using the Levene test.

For the variables feed and nutrient intake, ADG, feed efficiency, fecal score, ruminal fermentation variables, blood metabolites, and body measures, the analysis was performed as repeated measures over time (weeks of age), with the following statistical model: $Y_{ijk} = \mu + D_i + b_j + e_{ij} + A_k + (bA)_{jk} + (DA)_{ik} + e_{ijk}$, where μ = general average; D_i = fixed effect of diet; b_j = random block effect; e_{ij} = residual error; A_k = fixed age effect; $(bA)_{jk}$ = random effect of block × age interaction; $(DA)_{ik}$ = fixed effect of the diet × age interaction; and e_{ijk} = residual error B. The covariance matrices compound symmetry, heterogeneous compound symmetry, autoregressive, autoregressive heterogeneous, unstructured, banded, ante-dependence, variance components, Toeplitz, and heterogeneous Toeplitz were tested and defined according to the lowest value obtained for Akaike's information criterion corrected, and the subject of the repeated measures used was animal (treatment). For all the response variables, the means were obtained through the LSMEANS command. The comparisons among the treatments were performed by the Tukey test when there was significance in the ANOVA. Significance was declared when $P \leq 0.05$.

RESULTS

Intake and Growth Performance

During the preweaning period, SCH diet increased total DMI ($P = 0.01$; Table 2), as well as promoting the higher intake of CP and NFC ($P = 0.01$), although a progressive increase in intake occurred with age ($P < 0.01$). A significant interaction was observed between diet and age for NDF intake ($P < 0.01$; Figure 1), and the SCH diet promoted higher fiber intake from wk 4 until weaning. The SCH diet promoted higher starter concentrate intake ($P = 0.01$) than the other diets, and an increase in the intake of CP, NDF, and NFC from the starter concentrate ($P < 0.01$), with a significant effect of age ($P < 0.01$).

A significant interaction was observed between diet and age for total DMI postweaning ($P = 0.03$; Table 2), with the total DMI being higher at wk 9 ($P = 0.03$; Figure 1) for calves receiving the SCH diet. However, the SCW diet resulted in higher total DMI at wk 11 and 12 ($P = 0.02$; Figure 1). An interaction was observed between diet and age for NFC intake ($P = 0.04$). The NFC intake increased at wk 9 for calves fed SCH, and at wk 10 and 12, the SCW diet promoted the higher NFC intake. Diets did not alter starter concentrate intake during the postweaning period, although there was a progressive increase in intake with increasing age ($P < 0.01$).

During the preweaning period, SCH diet increased ADG ($P < 0.01$; Table 2); however, feed efficiency was not affected by diets. The BW at 21 d of age showed that calves started receiving whole-flint corn grain or hay, presenting the same weight averages. At weaning

(56 d), the SCH diet increased BW ($P = 0.01$). However, no effect of diet was observed for ADG, FE, or final BW postweaning.

The SCH diet increased withers height during preweaning ($P = 0.02$) and the hip width ($P = 0.01$; Table 3). No difference was observed in body measurements postweaning. However, withers height was affected by calves' age postweaning ($P = 0.01$). All body measures showed no interaction between treatment and age during the pre- or postweaning period.

Ruminal Variables

The SCW diet presented a lower molar concentration of acetic acid as compared with SCH, both pre- and postweaning ($P < 0.01$ and $P < 0.03$, respectively; Table 4). The SCW diet increased propionic acid and decreased acetate:propionate ratio ($P < 0.01$), as well as reduced the C2:C3+C4 ratio ($P = 0.01$) only preweaning. The butyric acid was unaffected by diets preweaning but increased for control calves postweaning. The concentrations of total VFA and NH_3-N were also not affected by diet pre- and postweaning. In contrast, ruminal pH was not affected preweaning, but it was higher for calves fed hay postweaning (SCH and SC).

Fecal Variables

The solid diets did not affect the fecal score or pH during the preweaning period. Postweaning, fecal score presented an interaction between diet and age ($P = 0.03$; Table 4), with increased fluidity of feces at wk 12 for calves fed SCW. Furthermore, the SCW diet resulted in lower fecal pH ($P = 0.03$) postweaning.

Table 2. Performance of calves receiving hay or whole corn grain free choice in the diet

	Diet[1]				*P*-value		
Item	SC	SCH	SCW	SEM	Diet	Age	D × A[2]
Solid feed intake, DM/d							
Preweaning, d 21–56							
Total solid diet, g/d	386.7[b]	635.8[a]	419.71[b]	50.75	0.01	<0.01	0.11
CP	95.7[b]	150.5[a]	84.8[b]	12.60	0.01	<0.01	0.06
NDF	54.7	112.6	56.8	7.45	<0.01	<0.01	0.01
NFC	181.5[b]	278.7[a]	221.3[ab]	24.30	0.01	<0.01	0.12
Starter concentrate, g/d	386.7[b]	595.8[a]	315.9[b]	46.53	0.01	<0.01	0.06
CP	95.7[ab]	145.1[a]	76.6[b]	12.50	0.01	<0.01	0.14
NDF	54.7[ab]	81.7[a]	43.4[b]	6.83	0.01	<0.01	0.09
NFC	181.5[ab]	273.9[a]	145.4[b]	22.92	0.02	<0.01	0.11
Postweaning, d 57–84							
Total solid diet, g/d	1,745.2	1,871.6	2,061.3	124.47	0.47	<0.01	0.03
CP	406.7	435.0	411.3	25.02	0.68	<0.01	0.16
NDF	358.0	396.4	275.1	28.60	0.02	<0.01	<0.01
NFC	717.3	778.3	1,111.0	60.75	0.01	<0.01	0.04
Starter concentrate, g/d	1,558.6	1,650.0	1,425.3	88.60	0.37	<0.01	0.50
CP	380.3	403.0	354.2	21.90	0.29	<0.01	0.48
NDF	214.1	226.5	199.7	10.63	0.08	<0.01	0.13
NFC	720.4	772.9	685.0	48.60	0.54	0.29	0.88
ADG, kg/d							
Preweaning, d 21–56	0.483[b]	0.647[a]	0.486[b]	0.038	<0.01	<0.01	0.51
Postweaning, d 57–84	0.913	0.850	0.969	0.078	0.44	0.72	0.30
Overall, d 21–84	0.598[b]	0.718[a]	0.634[ab]	0.036	0.02	<0.01	0.27
BW, kg							
Initial (d 21)	47.9	49.2	48.9	0.82	0.50	—	—
At weaning (d 56)	61.0[b]	68.9[a]	62.8[b]	1.82	0.01	—	—
Final (d 84)	87.3	89.8	91.9	2.85	0.52	—	—
Gain:feed ratio[3]							
Preweaning, d 21–56	0.52	0.60	0.52	0.037	0.16	0.16	0.55
Postweaning, d 57–84	0.46	0.43	0.42	0.046	0.83	0.06	0.20
Overall, d 21–84	0.53	0.56	0.50	0.027	0.15	0.03	0.62

[a,b]Means within a row with different superscripts are significantly different ($P \leq 0.05$).

[1]SC = starter concentrate only during preweaning and starter concentrate + free access chopped Tifton hay during postweaning; SCH = starter concentrate + free access chopped Tifton hay during pre- and postweaning; SCW = starter concentrate + free access whole-flint corn grain during pre- and postweaning; 14 calves per treatment.

[2]D × A = interaction between diet and age.

[3]Considering liquid and solid DMI.

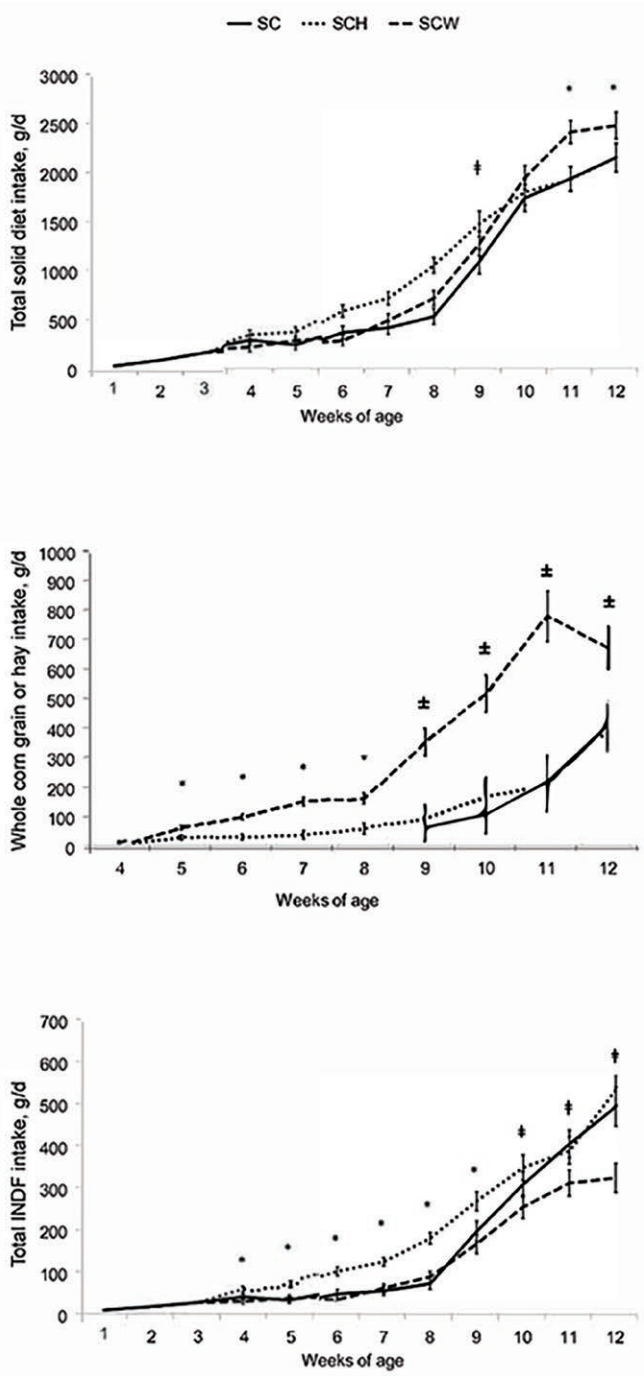

Figure 1. (A) Total solid diet intake (g/d). SC = starter concentrate only during preweaning and starter concentrate + free access chopped Tifton hay during postweaning; SCH = starter concentrate + free access chopped Tifton hay during pre- and postweaning; SCW = starter concentrate + free access whole corn grain during pre- and postweaning. ‡Denotes the difference between SCH and SC, with no difference from the SCW ($P = 0.03$). *Denotes difference between SCW than SCH and SC ($P = 0.02$). (B) Whole corn grain or hay intake (g/d). *Denotes the difference between SCW and SCH ($P = 0.01$). ±Denotes the difference between SCW than SCH and SC ($P < 0.01$). (C) Total NDF intake (g/d). *Denotes difference between SCH than SCW and SC ($P < 0.01$). ‡Denotes difference between SCH and SCW, with no difference from the SC ($P = 0.05$).

Blood Metabolites

During the preweaning period, the solid diet did not affect any of the selected blood variables. However, there was an age effect for all of them ($P < 0.02$; Table 5), and a diet and age interaction effect for BHB concentration ($P = 0.05$). The BHB concentration increased at wk 7 for calves fed the SCH diet ($P = 0.05$) compared with the SCW diet, with no difference from the SC diet. The plasma glucose increased in the first week's study and decreased wk 7 until the end of the study. Plasma lactate decreased as calves aged until weaning but increased after that. Serum protein concentrations decreased at wk 2 through wk 5 of life. As for plasma albumin, an increase occurred in concentration until wk 6, a decrease at wk 7, and subsequent stabilization at wk 8 of life. The plasma urea concentration increased until wk 6, decreased in wk 7, and increased in wk 8.

Postweaning, the SCW decreased lactate concentrations ($P = 0.03$; Table 5), as well as decreased urea concentration ($P < 0.01$). Also, regardless of diet, lactate concentration and TSP increased with age ($P = 0.01$). No significant interaction was observed between diet and age for all variables at the postweaning period.

Empty Forestomach and Organs

Calves fed with SCH diet presented the highest total weight (kg) of the proximal digestive tract ($P < 0.01$; Table 6). However, there were no significant differences in the proximal digestive tract as % of BW. The empty reticulum-rumen weight was higher in calves fed the SCH diet ($P = 0.03$) than the SCW diet without differing from the SC diet. No difference was observed among treatments for omasum and abomasum weight, reticulum-rumen volume, and small and large intestine weight.

Morphometric Variables

The diets did not affect the number of papillae per centimeter squared, height, width, papillae surface area, papillae KLT, and ASA (Figure 2). However, the papillae area expressed as % of ASA was higher in calves fed with SCH diet as compared with SCW, without differing from the SC diet ($P = 0.05$; Table 7). The depth of the crypts and the number of goblet cells from the duodenum were not affected by diets. However, the villus height of the duodenum was higher for animals that received SCH diet with no difference from the SC diet ($P = 0.02$; Figure 2). Diet did not alter the depth of the crypts and the number of goblet cells from the cecum.

DISCUSSION

Calf Intake and Growth Performance

In the past 2 decades, studies reporting the addition of forages in the diets observed negative effects when preweaning calves were fed free-choice high-quality forages, such as chopped alfalfa hay (17% CP and 47% NDF), with forage intake up to 14% of the total DMI (Terré et al., 2013b). The high palatability of this forage may depress the concentrate intake, resulting in a replacement effect, due to the low intake capacity of young calves (Beiranvand et al., 2014). Feeding high-quality forages tends to result in a higher forage: concentrate ratio than when the straw is fed (Castells et al., 2012).

In this study, the supply of tropical grass hay during the preweaning period increased total DMI and starter concentrate intake. Tropical grass hays, such as *Cynodon* (coast-cross, Tifton-85, and Florakirk, among others), have good nutritional value (11 to 14% CP), thin stems, and a large number of leaves, with good acceptance by ruminants (Poczynek et al., 2019). Poczynek et al. (2019) provided coast-cross hay ad libitum with nutritional value similar to the Tifton-85 hay used in this study (11.8 CP and 69.6 NDF) for milk-fed dairy calves; they reported a forage:concentrate ratio of 4:96, similar to the value observed in our study (7:93). The commercial starter concentrate used in the diets contained 13.8% NDF, slightly below the range recommended by Davis and Drackley (1998), who suggested levels between 15 and 25% NDF for calves. In these situations, tropical grass hay (74.7% NDF) free choice is a viable alternative in the diet of young calves with a developing rumen. Our study suggests that fiber from forage, when associated with starter concentrate with a high ruminal fermentation rate and fine particles that had been pelleted, may bring benefits such as concentrate intake stability and maintaining gain rates during the preweaning period.

The digestibility of nutrients from dry feeds (concentrates and forages) is dependent on the rumen and intestine development, mostly when these nutrients do not come from the liquid diet, which has higher digestibility. Thus, metabolic adaptations as a response to the ruminal event are required for proper digestion (Guilloteau et al., 2009). Cumulative concentrate NFC intake is highly correlated with changes in the apparent digestibility of the total tract, and an NFC value of 15 kg is required for digestive tract maturation (Quigley et al., 2019). The same authors reported that a cumulative rather than a daily intake of NFC was more related to changing apparent total-tract digestibility. These data suggest that the effects of carbohydrate fermentation in the rumen are cumulative; thus, promoting early and consistent starter intake might be advantageous during the transition phase. The increase in concentrate intake provided by free access to hay increased nutrient intake, such as CP and NFC, essential for maintenance and growth during the initial phase. The cumulative NFC intake during the preweaning period was higher for animals that received the SCH diet from 21 d (10.95

Table 3. Body measurement gains of calves receiving hay or whole corn grain free choice in the diet

	Diet[1]				*P*-value		
Item	SC	SCH	SCW	SEM	Diet	Age	D × A[2]
Preweaning, cm/wk, d 21–56							
Withers height	0.16[b]	0.24[a]	0.16[b]	0.020	0.02	0.27	0.58
Heart girth	0.05	0.08	0.07	0.010	0.09	0.18	0.15
Hip width	0.28	0.33	0.27	0.038	0.41	0.21	0.83
Preweaning, mean, d 21–56							
Withers height	84.49	85.91	84.84	0.62	0.17	<0.01	0.07
Heart girth	23.18	23.39	23.37	0.22	0.75	<0.01	0.45
Hip width	87.3[b]	90.0[a]	87.9[ab]	0.66	0.01	<0.01	0.81
Postweaning, cm/wk, d 57–84							
Withers height	0.23	0.21	0.25	0.036	0.76	0.01	0.73
Heart girth	0.08	0.06	0.08	0.012	0.39	0.58	0.79
Hip width	0.33	0.36	0.38	0.041	0.69	0.60	0.16
Postweaning, mean, d 57–84							
Withers height	90.70	91.97	90.66	0.95	0.52	0.01	0.60
Heart girth	25.93	25.71	25.38	0.25	0.90	<0.01	0.34
Hip width	100.1	100.5	99.9	1.43	0.95	<0.01	0.46

[a,b]Means within a row with different superscripts are significantly different ($P \leq 0.05$).

[1]SC = starter concentrate only during preweaning and starter concentrate + free access chopped Tifton hay during postweaning; SCH = starter concentrate + free access chopped Tifton hay during pre- and postweaning; SCW = starter concentrate + free access whole-flint corn grain during pre- and postweaning; 14 calves per treatment.

[2]D × A = interaction between diet and age.

± 1.120 kg) when compared with SC (7.80 ± 0.655 kg) or SCW (8.55 ± 0.869 kg). Although all groups showed cumulative intake below the levels suggested by Quigley et al. (2019), animals that received hay free choice showed the closest value. In addition, the animals fed SCH increased total DMI at wk 8, when the weaning process started (1,080.0 ± 78.21 g), compared with animals that received SC (564.0 ± 82.44 g) and SCW (746.0 ± 77.29 g), avoiding growth reductions during the transition phase and thereafter. These data suggest that calves fed grass hay preweaning were more prepared for weaning than those fed the other diets.

The SC treatment, which received hay after weaning, presented the same total DMI as the other treatments 1 wk after weaning (Figure 1). These data suggest that feeding calves grass hay before weaning is necessary to maintain feed intake and growth performance during the transition process when animals receive a commercial starter presented with highly fermentable and fine particles, as reported by Porter et al. (2007).

During the preweaning period, the SCH diet stimulated concentrate intake, resulting in higher ADG, without affecting FE. Thus, the increase in ADG is due to higher starter concentrate and, consequently, nutrient intake because of free access to hay, resulting in heavier animals at weaning. Mojahedi et al. (2018) observed an increase in ADG for animals that received hay free choice when associated with steam-flaked corn, suggesting that diets with higher ruminal starch digestibility or extensively processed grains require the addition of effective fiber. The data from the present study corroborate the study by Mojahedi et al. (2018). The effective fiber provided by feeding hay with the high NFC (46% DM) level starter, since the preweaning period, can better prepare the animal for the weaning period, improving ADG and BW.

Table 4. Ruminal fermentation and fecal variables of calves of calves receiving hay or whole corn grain free choice in the diet

	Diet[1]				*P*-value		
Item	SC	SCH	SCW	SEM	Diet	Age	D × A[2]
Preweaning, 6 and 8 wk							
Total, m*M*	129.2	146.5	128.8	12.29	0.44	0.31	0.82
VFA, m*M*/100 m*M*							
Acetate	48.9[ab]	52.0[a]	46.3[b]	1.04	<0.01	0.15	0.82
Propionate	36.7[ab]	32.9[b]	37.0[a]	1.27	0.02	0.50	0.37
Butyrate	8.1	7.8	8.8	0.59	0.46	0.12	0.21
Isobutyric	0.8	0.7	0.7	0.09	0.72	0.28	0.99
Valeric	3.9	3.5	4.2	0.31	0.38	0.21	0.27
Isovaleric	1.3	1.1	1.2	0.15	0.78	0.44	0.97
C2:C3 ratio[3]	1.3[ab]	1.6[b]	1.2[a]	0.06	<0.01	0.37	0.37
C2:C3+C4 ratio[3]	1.1[ab]	1.2[a]	0.9[b]	0.03	0.01	0.34	0.63
NH_3-N, mg/dL	16.6	12.9	17.2	1.80	0.09	0.13	0.88
Ruminal pH	5.4	5.5	5.4	0.12	0.70	0.49	0.84
Fecal score	1.5	1.6	1.4	0.10	0.67	0.01	0.81
Fecal pH at 2 h	7.1	6.9	7.1	0.09	0.12	0.95	0.50
Fecal pH at 7 h	6.8	6.9	6.7	0.16	0.88	0.10	0.34
Postweaning, 10 and 12 wk							
Total, m*M*	131.8	93.5	104.2	20.45	0.36	0.84	0.70
VFA, m*M*/100 m*M*							
Acetate	52.3[ab]	55.5[a]	45.2[b]	1.45	0.03	0.95	0.45
Propionate	34.9	33.6	41.4	2.30	0.15	0.42	0.74
Butyrate	8.1	5.7	5.8	0.55	<0.01	0.74	0.01
Isobutyric	0.4	0.8	0.5	0.12	0.17	0.24	0.05
Valeric	3.7	2.7	4.2	0.68	0.43	0.82	0.72
Isovaleric	0.7	0.8	0.8	0.11	0.73	0.35	0.43
C2:C3 ratio	1.5	1.7	1.2	0.11	0.06	0.26	0.60
C2:C3+C4 ratio	1.2[ab]	1.4[a]	0.9[b]	0.08	0.01	0.92	0.19
NH_3-N, mg/dL	20.0	22.2	20.7	4.60	0.89	0.16	0.37
Ruminal pH	6.1[a]	6.0[a]	5.5[b]	0.16	0.03	<0.01	0.11
Fecal score	1.3	1.4	1.4	0.06	0.40	0.01	0.03
Fecal pH, at 2 h	7.0[a]	6.9[a]	6.4[b]	0.10	<0.01	0.56	0.35
Fecal pH, at 7 h	6.8[a]	6.8[a]	6.4[b]	0.10	0.01	0.17	0.32

[a,b]Means within a row with different superscripts are significantly different ($P \leq 0.05$).

[1]SC = starter concentrate only during preweaning and starter concentrate + free access chopped Tifton hay during postweaning; SCH = starter concentrate + free access chopped Tifton hay during pre- and postweaning; SCW = starter concentrate + free access whole-flint corn grain during pre- and postweaning; 14 calves per treatment.

[2]D × A = interaction between diet and age.

[3]C2 = acetic acid; C3 = propionic acid; C4 = butyric acid.

Table 5. Blood metabolites of calves receiving hay or whole corn grain free choice in the diet

Item	Diet[1]			SEM	*P*-value		
	SC	SCH	SCW		Diet	Age	D × A[2]
Preweaning, d 21–56							
Glucose, mg/dL	114.5	109.2	115.1	3.37	0.47	<0.01	0.31
L-Lactate, mg/dL	8.7	8.1	8.2	0.60	0.75	0.01	0.79
BHB, mm/L	0.13	0.15	0.14	0.020	0.74	<0.01	0.05
Total protein, g/dL	5.8	6.1	5.8	0.11	0.06	<0.01	0.08
Albumin, mg/dL	3.1	3.1	3.1	0.04	0.42	0.01	0.87
Urea, mg/L	17.6	18.7	18.0	1.00	0.72	0.02	0.66
Creatinine mg/dL	1.1	1.1	1.1	0.02	0.70	0.02	0.11
Postweaning, d 57–84							
Glucose, mg/dL	82.1	84.4	85.6	2.10	0.50	0.39	0.99
L-Lactate, mg/dL	11.2[a]	11.0[a]	8.4[b]	0.72	0.03	0.01	0.95
BHB, mm/L	0.40	0.40	0.40	0.030	0.87	0.06	0.41
Total protein, g/dL	6.2	6.5	6.1	0.13	0.07	<0.01	0.30
Albumin, mg/dL	3.3	3.2	3.2	0.04	0.42	0.25	0.85
Urea, mg/L	32.3[a]	32.1[a]	19.5[b]	1.80	<0.01	0.07	0.12
Creatinine, mg/dL	1.0	1.0	1.0	0.03	0.58	0.07	0.37

[a,b]Means within a row with different superscripts are significantly different ($P \leq 0.05$).

[1]SC = starter concentrate only during preweaning and starter concentrate + free access chopped Tifton hay during postweaning; SCH = starter concentrate + free access chopped Tifton hay during pre- and postweaning; SCW = starter concentrate + free access whole-flint corn grain during pre- and postweaning. Fourteen calves per treatment.

[2]D × A = interaction between diet and age.

Imani et al. (2017) reported that the effects of forage provision to dairy calves on performance might also depend on the physical form of starter. The forage provision increased starter intake and ADG when the starter was ground or pelleted as compared with texturized. This effect was observed in the present study. However, whole-flint corn grain free choice did not increase starter intake and ADG as hypothesized.

However, the benefits observed preweaning seem to vanish postweaning, with similar ADG and FE among treatments. Free access to whole-flint corn grain postweaning provided higher total DMI and consequently higher NFC intake at wk 10 and 12, suggesting that more starch was available for ruminal fermentation and intestinal absorption, resulting in similar final BW (84 d). Mojahedi et al. (2018) reported that non-forage-fed

Table 6. Empty forestomach and organs weight of calves receiving hay or whole corn grain free choice in the diet at weaning

Item	Diet[1]			SEM	*P*-value
	SC	SCH	SCW		Diet
Forestomach, kg	1.23[b]	1.63[a]	1.23[b]	0.117	0.01
% BW	1.98	2.33	2.16	0.217	0.42
Reticulum-rumen, kg	0.79[ab]	1.10[a]	0.64[b]	0.100	0.03
% BW	61.75	66.17	61.28	3.470	0.57
Reticulum-rumen capacity (L)	9.19	10.60	8.75	0.890	0.43
Omasum, kg	0.101	0.170	0.290	0.080	0.40
% BW	11.43	12.70	11.59	0.860	0.54
Abomasum, kg	0.35	0.37	0.28	0.020	0.10
% BW	26.81	21.11	27.11	3.530	0.43
Small intestine, kg	2.24	2.54	2.07	0.160	0.21
Large intestine, kg	0.61	0.93	0.73	0.260	0.46
Kidney, kg	0.34	0.38	0.34	0.043	0.39
Liver, kg	1.65	1.90	1.62	0.104	0.19
Spleen, kg	0.32	0.44	0.60	0.102	0.27

[a,b]Means within a row with different superscripts are significantly different ($P \leq 0.05$).

[1]SC = starter concentrate only during preweaning and starter concentrate + free access chopped Tifton hay during postweaning; SCH = starter concentrate + free access chopped Tifton hay during pre- and postweaning; SCW = starter concentrate + free access whole-flint corn grain during pre- and postweaning; 4 calves per treatment; 8 wk of age.

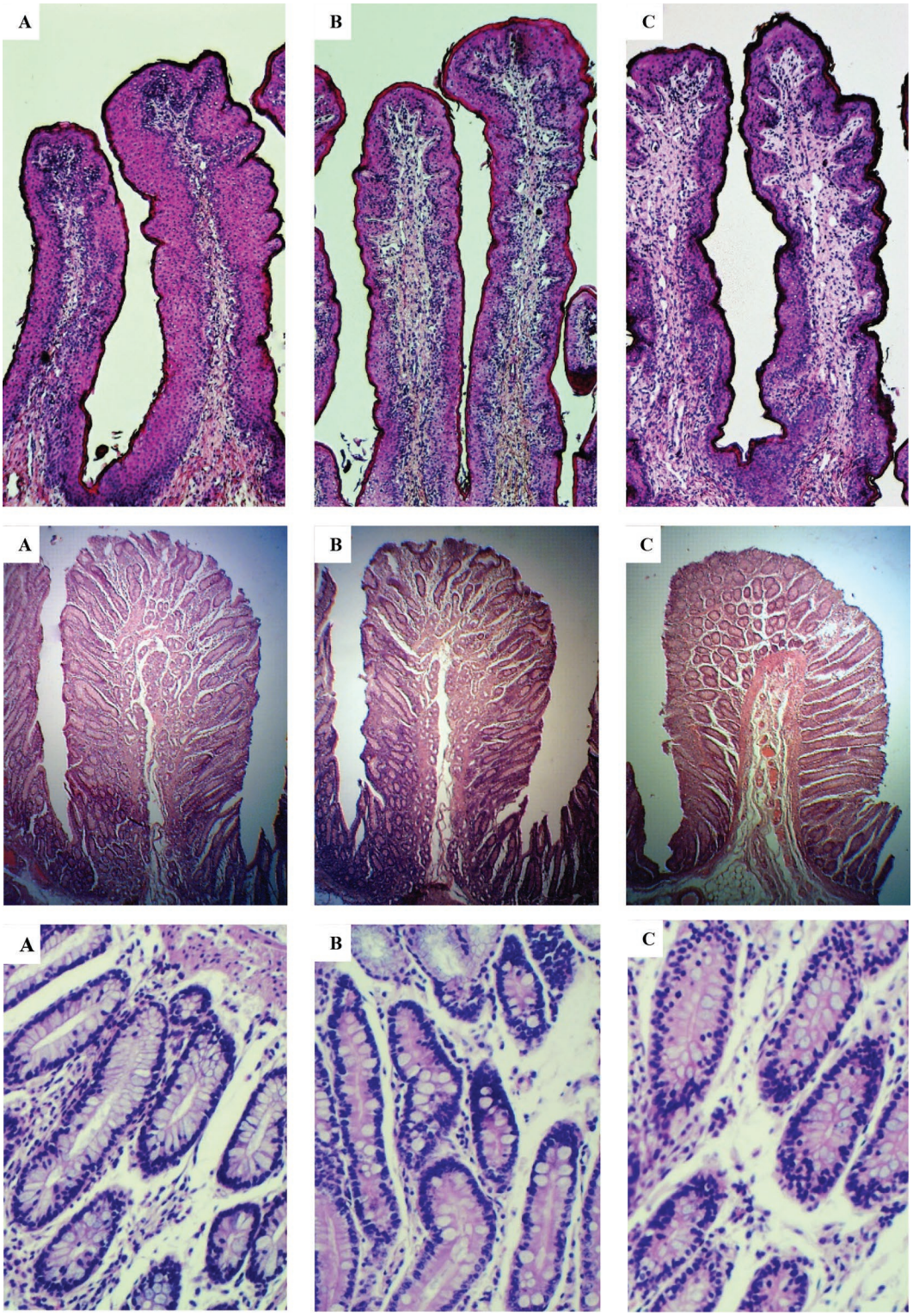

Figure 2. Electron micrographs from the rumen, duodenum, and cecum of calves at weaning receiving hay or whole corn grain free choice in the diet. (A) Starter concentrate only during preweaning and starter concentrate + free access chopped Tifton hay during postweaning; (B) starter concentrate + free access chopped Tifton hay during pre- and postweaning; and (C) starter concentrate + free access whole-flint corn grain during pre- and postweaning.

Table 7. Morphometric variables from the rumen, duodenum, and cecum at weaning of calves receiving hay or whole corn grain free choice in the diet

	Diet[1]				*P*-value
Item	SC	SCH	SCW	SEM	Diet
Rumen					
No. of papillae/cm^2	199.9	189.7	180.2	21.81	0.78
Papillae height, mm	0.98	1.20	0.95	0.131	0.35
Papillae width, mm	0.30	0.38	0.29	0.027	0.09
Papillae surface area, mm^2	0.30	0.45	0.29	0.060	0.17
Papillae KLT,[2] µm	11.42	12.45	11.76	0.384	0.21
ASA,[3] cm^2/cm^2 of ruminal wall	1.8	2.3	1.5	0.20	0.06
Papillae area, % of ASA	64.7^{ab}	72.8^{a}	54.4^{b}	4.25	0.05
Duodenum					
Villus length, µm	1,811.3^{ab}	2,051.2^{a}	1,287.1^{b}	195.30	0.02
Crypt depth, µm	83.7	81.9	84.9	13.13	0.99
No. of goblet cells/crypt	5.7	6.5	3.5	1.08	0.18
Cecum					
Crypt depth, µm	102.1	108.2	112.5	11.21	0.81
No. of goblet cells/crypt	9.0	11.6	8.7	1.37	0.27

[a,b]Means within a row with different superscripts are significantly different ($P \leq 0.05$).

[1]SC = starter concentrate only during preweaning and starter concentrate + free access chopped Tifton hay during postweaning; SCH = starter concentrate + free access chopped Tifton hay during pre- and postweaning; SCW = starter concentrate + free access whole corn grain during pre- and postweaning; 4 calves per treatment.

[2]KLT = keratinized layer thickness.

[3]ASA = absorptive surface area.

calves had a sharp increase in BW as compared with forage-supplemented calves. Still, the potential mechanisms that elicit compensatory growth improvements were not clear. Consistent with our results, the effects of forage supply or grain processing had minor or no effects on skeletal growth variables, as reported in other studies (Mirzaei et al., 2016; Mojahedi et al., 2018). However, free-choice hay positively affected withers height, demonstrating that the animals that presented greater weight also grew more during the preweaning phase. Animals that gain weight but do not show body size changes may have unwanted gain composition, such as adipose tissue deposition rather than muscle tissue (Heinrichs et al., 2003).

Ruminal Variables

Although the diets changed the ruminal fermentative profile during the pre- and postweaning phases, total VFA concentrations were not affected by diet or age. However, calves were sampled only twice in each evaluated period. This increase in total VFA concentrations is due to the increase in the fermentation capacity observed with advancing age and, consequently, ruminal development (Nussio et al., 2003).

Feeding SC and SCW diet increased propionic acid concentrations and provided a lower acetate:propionate ratio during the preweaning period, a common effect when the high starch diet is fed (Theurer, 1986; McGuffey et al., 2001). On the other hand, free-choice tropical grass hay increased acetic acid concentration in both periods. This increase in the molar proportions of acetic acid has been reported in hay-added diets for calves due to changes in the ruminal microbiota (Mirzaei et al., 2016; Nemati et al., 2016). Animals that received SC and SCW diets had a lower C2:(C3 + C4), indicating a higher concentration of propionic and butyric acids, responsible for stimulating ruminal development with regard to number, height, and width of papillae (Khan et al., 2016; Poczynek et al., 2019). However, the increased C2:(C3 + C4) was not accompanied by increased concentrate intake, which is the main factor for ruminal development.

Animals fed an SC diet, and with free access to hay after weaning, presented higher butyrate concentrations at wk 10 of life. Butyrate concentrations in the rumen are lower during preweaning and increase as concentrate intake increases (Kosiorowska et al., 2011). However, concentrate intake did not increase during the postweaning period.

During the preweaning period (sampling at wk 6 and 8 of age), the ruminal pH was 5.5 ± 0.12, regardless of the experimental diet. In adult cattle, this value indicates of SARA and can lead to metabolic disorders (Plaizier et al., 2012). The supply of hay or fibrous materials such as by-products can increase and stabilize the ruminal pH. These ingredients have fiber that stimulates rumination and salivation (McBurney et al., 1983), buffering the total tract, a significant factor for the calf's adequate ruminal development and general

health (Khan et al., 2011; Castells et al., 2012, 2013). On the other hand, whole grains added to textured feeds may be more effective in supporting ruminal health and animal performance than highly processed grains (Terré et al., 2015). However, there were no positive effects of free access to whole-flint corn grain during the preweaning phase in total diet intake and ruminal pH after weaning. Ruminal pH was higher for the SCH diet, demonstrating that after weaning tropical grass hay has a positive effect on ruminal pH control. However, rejecting the study hypothesis, whole-flint corn grain free choice had no beneficial effect on intake, gain, or ruminal pH. It probably did not promote the level of physical effects required to offset the starch provided by the whole corn and starter.

Fecal Variables

Whole corn grain resulted in more liquid stools during the last week of the study, probably because of the NFC intake compared with diets containing the hay. Higher consumption of whole corn grain may have provided a greater corn escape from the rumen, increasing fermentation in the small intestine and later in the large intestine, which may cause cecal acidosis. This assumption can be reinforced by lower fecal pH in animals fed the SCW diet (Table 4). Thus, animals with free access to whole corn grain, leading to a reduction in ruminal and cecal pH postweaning, may have suffered adverse effects, such as metabolic disorders, which may decrease performance in the longer term. Although the ADG of the animals was not different after weaning, these data suggest that in a longer-term assessment, animals with free access to whole-flint corn grain would decrease their performance.

Blood Metabolites

The plasma glucose values in this study follow literature for calves of similar age fed whole milk (Mojahedi et al., 2018). The reduction in glucose concentrations with the advancing age of calves occurs regardless of the diet and is related to ruminal development (Quigley et al., 1992). According to Khan et al. (2008), a reduction in glucose concentrations is linked to a greater solid diet intake, which stimulates fermentation and ruminal activity. The VFA produced by the fermentation process gradually begins to replace glucose, becoming the primary energy source for animals (Baldwin et al., 2004). In this study, experimental diets did not affect plasma glucose concentrations pre- or postweaning; however, a reduction occurred in glucose concentrations from wk 7 onward.

On the other hand, BHB concentrations were not different among diets during the preweaning phase. However, BHB concentration was higher for animals that received SCH as compared with the SCW diet at wk 7, probably due to the higher concentrate intake.

In this study, contrary to expectations, the SCW diet decreased postweaning lactate concentration, possibly because, after weaning, the microbial population was more adapted to NFC degradation. However, no increase occurred in the propionic acid concentrations, and ruminal pH was lower in animals that received whole-flint corn grain corn.

The plasma urea concentration may reflect the ruminal microbial community composition and ammonia nitrogen levels, positively related to the concentrate intake (Eicher-Pruiett et al., 1992). Whole-flint corn may have increased the availability of carbonic skeletons for bacteria, increasing the use for microbial protein synthesis and reducing plasma urea concentrations in the postweaning period.

Empty Forestomach and Organs

Calves fed an SCH diet had a higher weight of the reticulum-rumen and greater total weight of the forestomach at the weaning week. The particle size from roughage provides physical stimuli to increase ruminal motility, muscularization, and ruminal volume in calves (Khan et al., 2016). On the other hand, when analyzing the weight of the compartment as a percentage of the calf BW, no effect was observed of the different solid diets for the analyzed variables, as well as for the ruminal volume. These results indicate that the higher weight of compartments for animals fed SCH diet may be due to their higher BW and not due to the increase in the ruminal musculature in a higher developmental stage, as reported in the literature (Tamate et al., 1962; Khan et al., 2011; Kosiorowska et al., 2011).

Lesmeister and Heinrichs (2004) reported that the transition from a preruminant calf to a fully functional ruminant with minimal loss in growth requires adequate size and development of the reticulum-rumen for efficient use of dry and forage-based diets. The literature collectively demonstrates that forage provision positively affects rumen functions and ruminal pH in calves; however, available studies are inconclusive in defining the exact amount, source, method, and timing of forage provision to calves (Khan et al., 2016; Imani et al., 2017). In our study, free-choice tropical grass hay, when associated with the highly fermentable and fine particle starter since 21 d of age, improved weight of the empty forestomach and reticulum-rumen, ensuring physical and functional development to meet the

demands of consuming fermentable carbohydrates and forage after weaning.

The provision of high levels of forage (>10% in DM) could increase gut fill during the preweaning and overall periods, reducing starter intake and, consequently, the energy density of the diet (Imani et al., 2017). Increased gut fill could be a confounding factor when determining forage effects on improving ADG and BW of dairy calves (Hill et al., 2008; Khan et al., 2011; Mirzaei et al., 2016). Although the supply of free-choice hay increased the weight of the empty forestomach, intestinal tissues were not changed by experimental diets, suggesting that lower levels of forage (≤10% of the diet DM) did not affect gut fill in the same way compared with higher levels of forage (>10% of the diet DM) fed to calves.

Ruminal Wall, Duodenum, and Cecum Morphometrics

Animals fed an SCH diet in the preweaning period presented a greater papillae area expressed as % of ASA as compared with the SCW diet, with no difference from SC (Figure 2). This result can be explained by the higher concentration of plasma BHB at wk 7 of life in calves that received hay free choice. The increase in blood flow through the ruminal wall (Sander et al., 1959), along with the direct effect of butyrate and propionate, are proposed as the main factors to trigger differentiation of the papillae. Butyrate is believed to be the main VFA to stimulate the development of ruminal epithelium (Flatt et al., 1958; Sander et al., 1959; Baldwin et al., 2004). Thus, tropical grass hay, which resulted in higher starter concentrate intake during the preweaning phase, may have provided more substrate for fermentation, promoting more efficient ruminal development as shown by others (Khan et al., 2011; Castells et al., 2013). Also, forage supply contributes to ruminal development by increasing its physical capacity, stimulating motility, and reducing parakeratosis, which is associated with high grain intake and acid accumulation in the rumen (Laarman et al., 2012; Khan et al., 2016).

During the experimental period, feeding tropical grass hay free choice caused no substitution effect on concentrate intake that could be unfavorable to the development of the gastrointestinal tract. We expected that whole grain of vitreous corn, rich in vitreous endosperm, compact, with a dense protein matrix, and with less starch available, could reduce secondary fermentation in the small intestine and cecum. The nonprocessed flint corn could reduce stress on the intestinal epithelium caused by the accumulation of acids and LPS, as reported by Kim et al. (2011). However, the SCW diet decreased the duodenum villi height (Figure 2), indicating that lower nutrient intake hindered the development of the initial portion of the intestine. In the presence of nutrients, the segment's digestive and absorptive capacity is directly proportional to the number and size of villi available for absorption (Steinhoff-Wagner et al., 2014). However, cecum development was not affected due to the different solid diets in the present study.

CONCLUSIONS

Feeding a chopped tropical grass hay free choice with a high NFC and fine particle pelleted starter concentrate increased preweaning starter concentrate and total solid intake and improved ruminal and intestinal development weaning, resulting in higher growth performance during the transition period. On the other hand, feeding whole-flint corn grain free choice did not increase nutrient intake and performance, pre- and postweaning, probably because of its negative effect on ruminal fermentation and its consequences on ruminal and intestinal development. In diets with overprocessed corn grain and high NFC, more effective fiber from forage is required to maintain feed intake and growth performance and positively aid in the ruminal wall and intestine development. Despite calves being housed outdoors in shelters distributed over a grass field, herbage was trimmed regularly to ground level and animals had no access to vegetative grass for grazing.

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